



SRI SATHYA SAI INSTITUTE OF HIGHER LEARNING
(Deemed to be University)
Department of Mathematics and Computer Science

CIE - II

Data Preparation · Principles of Data Wrangling

UDSC-402: Data Analysis and Visualization

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I. State whether the following statements are True or False. [3 × 1 = 3]

1. Data leakage can occur even when the rows of the test set are never directly used to fit the model.
2. In the Interquartile Range (IQR) method of outlier detection, the factor $k = 1.5$ is used to identify “extreme outliers” or “far outs”.
3. Schema-on-read means that data must adhere to structural and syntactic constraints in order to be ingested.

II. Fill in the blanks. [3 × 1 = 3]

4. A column in which every row contains the exact same value carries no information for modelling and is called a _____ predictor.
5. Iterative Imputation, in which each feature with missing values is modelled as a function of the other features and the process is repeated, is also known as _____.
6. A sample that provides representation of every group or “stratum” present in the data — though it may bias overall trends by over-representing small groups — is called a _____ sample.

III. Match the following data transforms with their correct definitions. [4 × 1 = 4]

Column A — Transform	Column B — Definition
i. Normalization Transform	a. Encode a categorical variable into binary (dummy) variables
ii. One Hot Transform	b. Change the distribution of a variable to be more Gaussian
iii. Discretization Transform	c. Scale a variable to the range 0 to 1
iv. Power Transform	d. Encode a numeric variable as an ordinal variable

Q7. List the FIVE core data preparation tasks in a predictive modelling project and give a one-line explanation of each. State which task is normally performed first and why. [5 Marks]

Q8. Explain the FIVE characteristics used to describe data in the data-wrangling framework. Answer in points, with one example for each. [5 Marks]

Q9. A student builds a classifier as follows: he first scales the ENTIRE dataset using MinMaxScaler, then splits it 67:33 into train and test, fits the model and reports 98% accuracy. In deployment, the model performs far worse. [5 Marks]

(a) Name the problem and explain why it arises here.

- (b) State the correct sequence of steps for a train-test split.
- (c) State how the same problem is avoided under k-fold cross-validation.

Q10. While profiling the FEC Candidate Master File you find: (i) Column 14 (state of the candidate's mailing address) has 57 distinct values, including the value “ZZ” and some missing entries; and (ii) Column 4 (election year registered for) ranges from 1990 to 2064, although your project concerns only the 2016 presidential election. [5 Marks]

- (a) Identify which type of profiling detects each issue.
- (b) Explain why 57 values is a problem for Column 14.
- (c) State the recommended action for the out-of-scope election years and justify it.